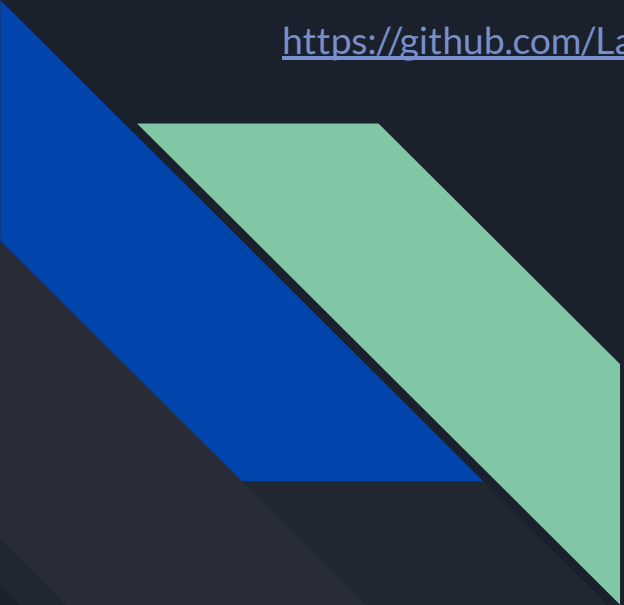




WasteLesss

“Smarter Inventory, Less Waste”

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What is WasteLesss?

Introduction & Project Motivation

WasteLesss addresses critical inventory management challenges businesses face:

- High costs due to holding excess inventory
- Losses from perishable items expiring
- Difficulty tracking inventory efficiently using outdated methods
- Financial loss from inventory mismanagement and dissatisfied customers

Our solution provides:

- Real-time inventory tracking
- Customizable item profiles
- Aging inventory alerts
- Demand forecasting to improve inventory decisions

The result: reduced waste, lower operational costs, improved decision-making, and enhanced satisfaction for customers and employees.

Who is WasteLesss For?

It is always a challenge when dealing with perishable goods. I would like an easy way to deal with products that are close to expiration date

Keeping track of my inventory is very hard. I don't just sell items, I use ingredients and if I run out of something, I can't serve certain dishes.

The problem is that without a good system, things get lost, and we don't always know what's close to expiring until it's too late. It would help if we had a way to track what is going in and out in real-time and get alerts before things expire.

I face dissatisfied customers because of lack of inventory information. There are times when a customer will see a product online, then come to the store to buy it but they cannot find it on the shelf.

Grocery Store
Manager

Restaurant
Owner

Warehouse
Manager

Grocery Store
Employee



Challenges with Inventory Management



Handling Inventory Costs

Customers are facing challenges in managing inventory costs, especially with obsolescence and perishables. They seek effective strategies to reduce losses while optimizing stock levels.



Inventory Tracking Basics

Many are struggling with tracking inventory accurately, particularly the time items spend in storage. This includes both perishables and non-perishables, leading to potential waste.



Addressing Time Intervals

Businesses need to address the lack of specific time intervals for inventory alerts. This oversight can result in aged or lost products, impacting overall efficiency.



Reducing Overstock Waste

Overstocking and poor demand forecasting lead to significant waste. Companies must find ways to minimize excess inventory and reduce losses from expired products.

WasteLesss'
Goal
2021

To improve inventory efficiency, reduce waste, reduce profit loss and improve decision making the system will provide real time tracking and notification with alternative methods of inventory handling.



Summarization of System Specifications

Specification Type	Description
Functional	Customizable Customer Data • Enable customers to add product specifications tailored to their needs. • Allow customer classification of aging criteria Real-time System Execution • Real-time data validation and reporting wich links inventory with customer sales data or ERP software to accurately track incoming and outgoing inventory • Provide real-time access to stock levels and age across various locations • Provide detailed reports on aging stock Inventory Management methods • Implement batch tracking • Use industry standards of track such as FIFO Notifications/ Alerts • Provide alert notifications • Prioritize alert notifications of older or aging inventory



Summarization of System Specifications

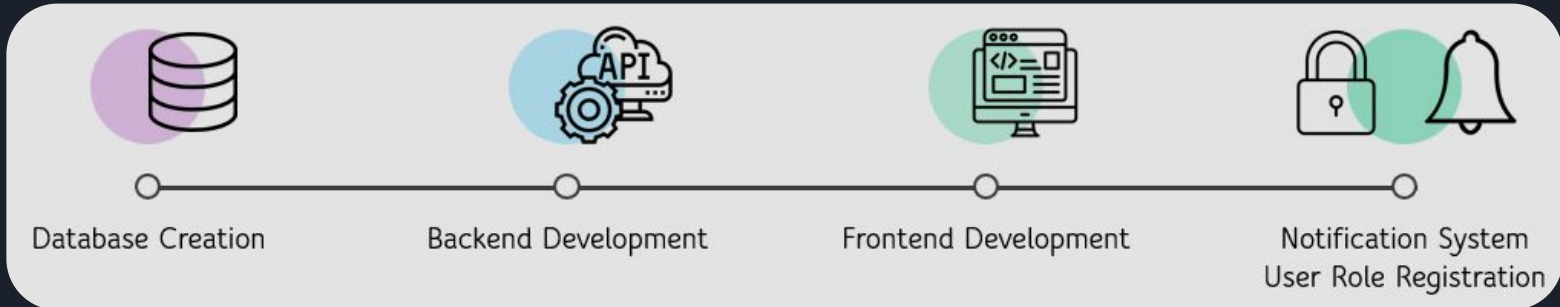
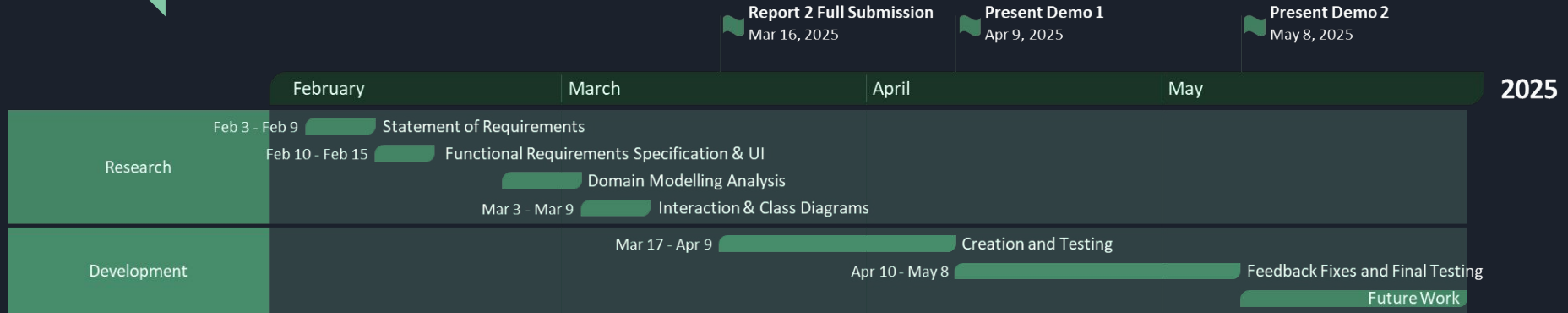
Specification Type	Description
Non-Functional	• Easy navigation and customer usability • Ability to identify problems directly in system • Allow for the creation of new user accounts with specific privileges • Support a high number of concurrent users • Allow for the user to add different suppliers • User should not have to wait long periods to get results • Inventory updates immediately to allow for real time tracking. • Software should be designed in a modular way
User Interface	Navigation Dashboard Overview • Separation of Use Case Inventory Management Interface (CRUD REST API) • Enter Inventory • Retrieve Inventory • Update Inventory • Delete Inventory Alerts & Notifications • Choose Method of notifications



Summarization of Programming Languages

Component	Technology/Tool Used
User Interface	HTML, CSS, JavaScript
Backend	Node.js, Express.js
Database	PostgreSQL
API Communication	RESTful API, CRUD Ops
Version Control	Git, GitHub
Optional Tools	Postman (for API testing)

Product Road Map





Project Challenges and Future Work

Project Challenges

- Faced challenges aligning frontend and backend connections
- User authentication and role registration
- Difficulty implementing and syncing alert functions across the system, ensuring they triggered accurately and at the right times
- Collaborating across multiple time zones and diverse programming languages

Integration & Interoperability

- Improve email notification system
- Connect with ERP/CRM systems (SAP, Salesforce)
- Add supplier portal for inventory & pricing updates

Data & Intelligence

- Implement stock checks before orders
- Create analytics dashboards for trends & supplier performance
- Use ML for cost optimization & demand forecasting

User Experience

- Improvements to seamlessly intertwine the email notification system with the GUI
- In-app tips and learning modules
- Suggestion voting for feedback



Conclusion

- WasteLess is built to tackle real inventory problems with a focus on efficiency, accuracy, and simplicity.
- We've listened to our users and created a strong foundation that meets their everyday needs.
- Looking ahead, we're excited to keep improving—with smarter integrations, better analytics, and a smoother user experience.
- WasteLess isn't just solving today's problems—it's built to grow with the businesses of tomorrow.